

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 6]

At a café, the waiting time between ordering and receiving a cup of coffee is dependent upon the number of customers who have already ordered their coffee and are waiting to receive it.

Sarah, a regular customer, visited the café on five consecutive days. The following table shows the number of customers, x , ahead of Sarah who have already ordered and are waiting to receive their coffee and Sarah's waiting time, y minutes.

Number of customers (x)	3	9	11	10	5
Sarah's waiting time (y)	6	10	12	11	6

The relationship between x and y can be modelled by the regression line of y on x with equation $y = ax + b$.

- (a) (i) Find the value of a and the value of b . [3]
- (ii) Write down the value of Pearson's product-moment correlation coefficient, r . [3]
- (b) Interpret, in context, the value of a found in part (a)(i). [1]

On another day, Sarah visits the café to order a coffee. Seven customers have already ordered their coffee and are waiting to receive it.

- (c) Use the result from part (a)(i) to estimate Sarah's waiting time to receive her coffee. [2]

.....

.....

.....

.....

.....

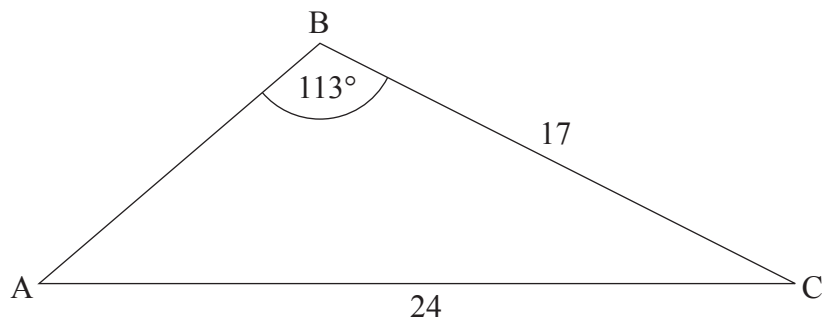
.....



2. [Maximum mark: 6]

The following diagram shows triangle ABC , with $AC = 24$, $BC = 17$, and $\hat{A}BC = 113^\circ$.

diagram not to scale



(a) Find $\hat{B}AC$. [3]

(b) Find AB . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

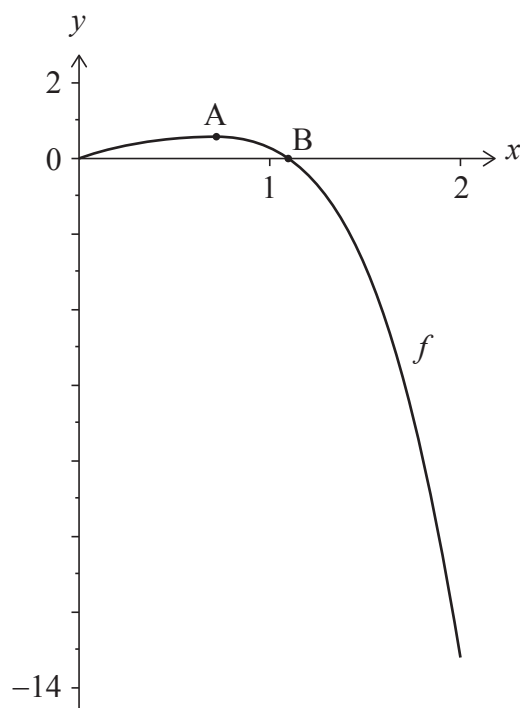
.....

.....



3. [Maximum mark: 6]

The function f is defined as $f(x) = \ln(xe^x + 1) - x^4$, for $0 \leq x \leq 2$. The graph of f is shown in the following diagram.



The graph of f has a local maximum at point A. The graph intersects the x -axis at the origin and at point B.

- Find the coordinates of A. [2]
- Find the x -coordinate of B. [1]
- Find the total area enclosed by the graph of f , the x -axis and the line $x = 2$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



4. [Maximum mark: 5]

A geometric sequence has a first term of 50 and a fourth term of 86.4.

The sum of the first n terms of the sequence is S_n .

Find the smallest value of n such that $S_n > 33\,500$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



5. [Maximum mark: 7]

The population of a town t years after 1 January 2014 can be modelled by the function

$$P(t) = 15\,000e^{kt}, \text{ where } k < 0 \text{ and } t \geq 0.$$

It is known that between 1 January 2014 and 1 January 2022 the population decreased by 11 %.

Use this model to estimate the population of this town on 1 January 2041.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



6. [Maximum mark: 6]

Consider the expansion of $\frac{(ax+1)^9}{21x^2}$, where $a \neq 0$. The coefficient of the term in x^4 is $\frac{8}{7}a^5$.

Find the value of a .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



Do **not** write solutions on this page.

8. [Maximum mark: 16]

The function f is defined by $f(x) = \frac{4x+1}{x+4}$, where $x \in \mathbb{R}$, $x \neq -4$.

- (a) For the graph of f
- (i) write down the equation of the vertical asymptote;
 - (ii) find the equation of the horizontal asymptote. [3]
- (b) (i) Find $f^{-1}(x)$.
- (ii) Using an algebraic approach, show that the graph of f^{-1} is obtained by a reflection of the graph of f in the y -axis followed by a reflection in the x -axis. [8]

The graphs of f and f^{-1} intersect at $x = p$ and $x = q$, where $p < q$.

- (c) (i) Find the value of p and the value of q .
- (ii) Hence, find the area enclosed by the graph of f and the graph of f^{-1} . [5]



Do **not** write solutions on this page.

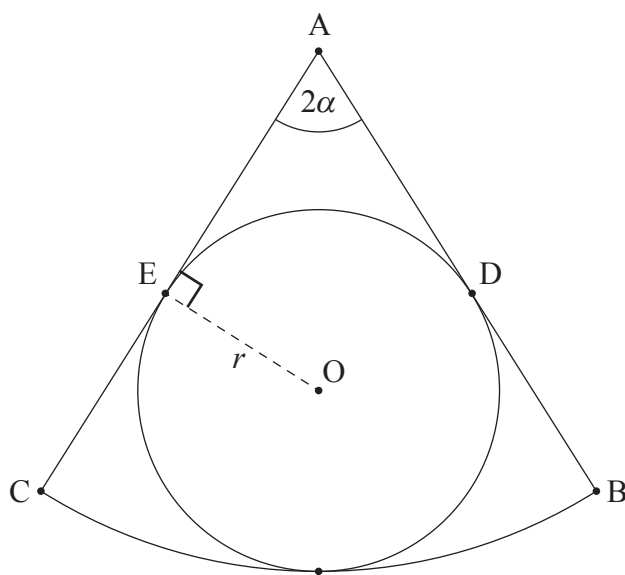
8. [Maximum mark: 13]

The following diagram shows a sector ABC of a circle with centre A . The angle $\widehat{BAC} = 2\alpha$, where $0 < \alpha < \frac{\pi}{2}$, and $\widehat{OEA} = \frac{\pi}{2}$.

A circle with centre O and radius r is inscribed in sector ABC .

AB and AC are both tangent to the circle at points D and E respectively.

diagram not to scale



(a) Show that the area of the quadrilateral $ADOE$ is $\frac{r^2}{\tan \alpha}$.

[4]

(This question continues on the following page)

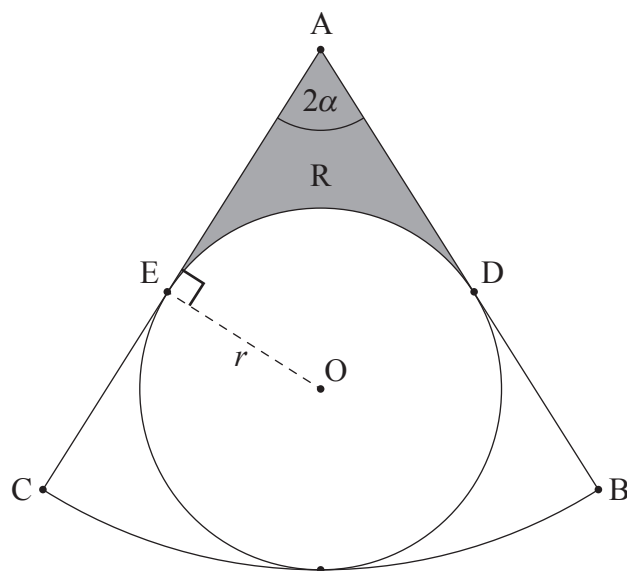


Do **not** write solutions on this page.

(Question 8 continued)

R represents the shaded region shown in the following diagram.

diagram not to scale



- (b) (i) Find $\angle DOE$ in terms of α .
- (ii) Hence or otherwise, find an expression for the area of R. [5]
- (c) Find the value of α for which the area of R is equal to the area of the circle of centre O and radius r . [4]



Do **not** write solutions on this page.

9. [Maximum mark: 16]

The time worked, T , in hours per week by employees of a large company is normally distributed with a mean of 42 and standard deviation 10.7.

(a) Find the probability that an employee selected at random works more than 40 hours per week. [2]

(b) A group of four employees is selected at random. Each employee is asked in turn whether they work more than 40 hours per week. Find the probability that the fourth employee is the only one in the group who works more than 40 hours per week. [3]

(c) A large group of employees work more than 40 hours per week.

(i) An employee is selected at random from this large group.

Find the probability that this employee works less than 55 hours per week.

(ii) Ten employees are selected at random from this large group.

Find the probability that exactly five of them work less than 55 hours per week. [7]

It is known that $P(a \leq T \leq b) = 0.904$ and that $P(T > b) = 2P(T < a)$, where a and b are numbers of hours worked per week. An employee who works fewer than a hours per week is considered to be a part-time employee.

(d) Find the maximum time, in hours per week, that an employee can work and still be considered part-time. [4]

References:

© International Baccalaureate Organization 2022

